

AI gets physical

The decade of the robot belongs to China

China's robotics roll-out is unmatched, accounting for 50% of global industrial robots and 85% of humanoids in 2025. Backed by coordinated industrial policy and tight supply chain control, humanoids could reach 3.8% of labour capacity by 2035, offsetting c.60% of China's projected workforce decline.



- **China is the world's robotics powerhouse and innovation lab**, installing around half of all industrial robots globally (nearly 300k versus just 34k in the US), lifting robot density by 600% to nearly 500 robots per 10k workers since 2016. It also now dominates the production and deployment of **humanoid robots**, accounting for 85% of global installations in 2025.
- **This leadership reflects a decade-long, state-guided push**. Our analysis of over 300k robotics patents since 2000 reveals that China's innovation has scaled at a remarkable pace, accounting for around 70% of global filings, versus 4% in the US. China's robotics engine is anchored in public research, with universities and state institutes accounting for 30% of inventions, compared with 10% in the US.
- **In China, robotics is industrial policy**. Policy spans the full stack – from critical inputs and energy through components and deployment – simultaneously shaping supply and creating domestic demand. China now dominates the humanoid supply chain end-to-end, with over

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⁽ⁱ⁾ This author is a debt research analyst in the Fixed Income, Currencies and Commodities Research department and is neither an equity research analyst nor subject to all of the independence and disclosure standards applicable to analysts who produce debt research reports under U.S. FINRA Rule 2242.

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90% of global refined rare earths. It also leads in key components, such as batteries, where it holds the largest global export share at 45%.

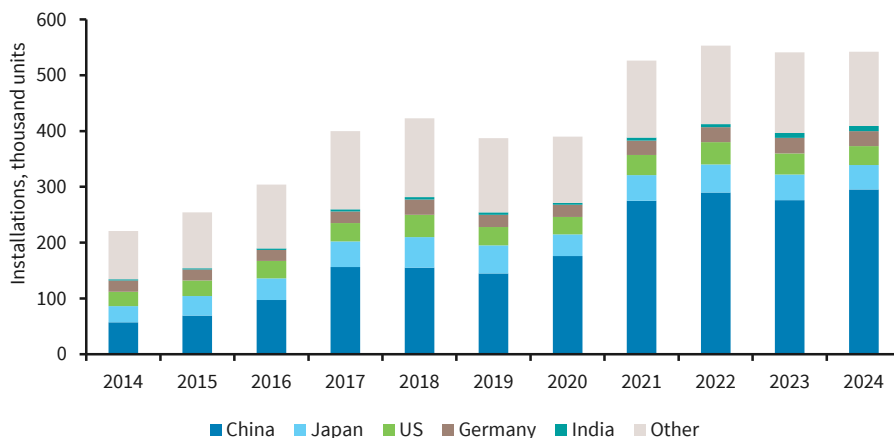
- **This is the decade of the robot – and it belongs to China.** Large-scale humanoid deployment began in 2025 with 15k units, rising to 60k in 2026 based on announced projects. Drawing on the roll-out of comparable technologies – most notably EVs in China – we project that by 2035 annual deployments could reach up to 13mn units as an upper bound, with roughly 11mn in China and 2mn elsewhere, primarily in the US.
- **China's rapidly aging population is driving strong domestic demand for robots.** A 37mn labour shortfall this decade is straining its \$4.7trn manufacturing base, while creating a vast domestic market to absorb millions of robots. If current trends hold, humanoid robots could add up to 3.8% to China's effective labour force by 2035 – roughly a stock of 24mn units – helping to offset up to 60% of the projected workforce decline by filling demographic gaps.
- **China has the edge – but the US and Europe are still in the race.** Adoption in Western markets is likely to come later, reflecting a more consumer-led model. However, security concerns around Chinese humanoids – such as risks to data and intellectual property – are likely to impose guardrails on their deployment. This creates scope for Western producers to scale through consolidation and industrial policy support.
- **Autonomy meets geopolitics and industrial policy.** Roughly one-fifth of robotics patents are linked to the defence industry, with further spillovers into aerospace, mobility and energy – underpinned by shared foundational technologies such as sensing and navigation. The takeaway for investors and policymakers is clear: **strategic value lies in the technology stack**, not just the end product, elevating the robotics supply chain to critical infrastructure.

The year of the robot

Automation in China started with industrial robots

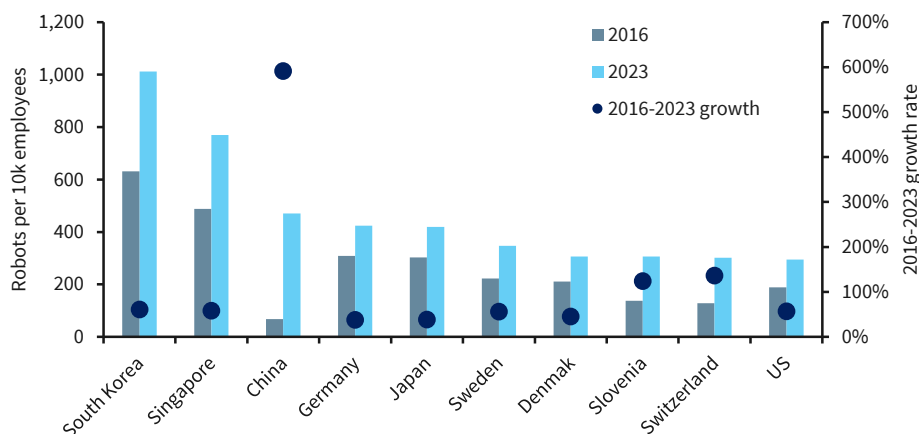
Robotics integration in China has advanced steadily since the early 2000s, unfolding in successive adoption waves. Under sustained state guidance, the first major automation push centered on industrial robots, with uptake accelerating sharply in the mid-2010s and scaling rapidly thereafter. Today, **China accounts for roughly half of global annual industrial-robot installations**, deploying nearly 300k units in 2024, with Japan and the US trailing far behind at 44k and 34k units ([Figure 1](#)).¹

¹ See International Federation of Robotics' [press release](#).

FIGURE 1. One in every two industrial robots is installed in China

Source: International Federation of Robotics, Barclays Research

The scale of growth is most pronounced in manufacturing. **Between 2016 and 2023, China lifted robot density by more than 600%:** from around 70 to nearly 500 robots per 10k workers, the fastest growth rate globally (Figure 2).²

FIGURE 2. China's manufacturing robot density is up 6x in a decade

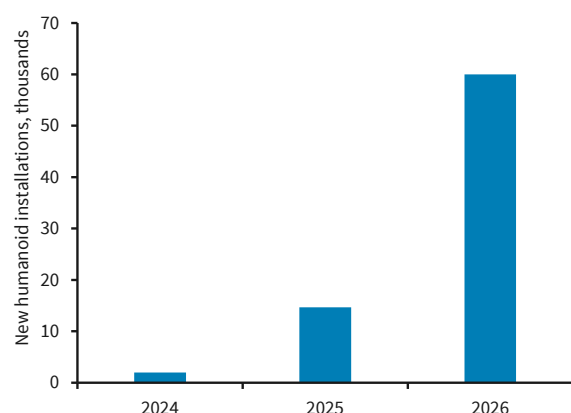
Source: International Federation of Robotics, Barclays Research

Automation's next act will be humanoids

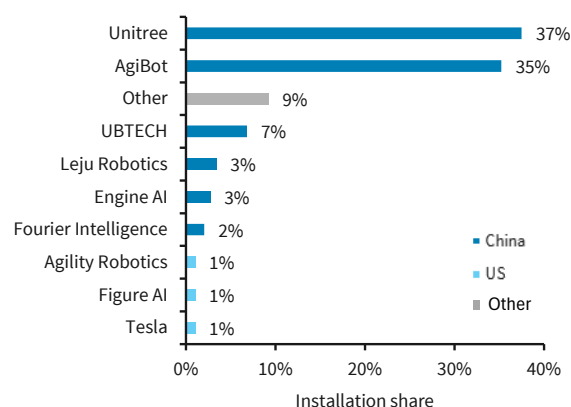
In our view, AI-enabled robotics – humanoids in particular – will be automation's next act, shifting automation from task-specific, factory-bound machines to systems designed to operate within human-built environments, using the same tools, layouts and workflows as people, and in doing so widening the automation frontier.

Although humanoid robots have existed for nearly a decade, **2025 marked a clear inflection point in deployment.** Based on our analysis of company announcements and press releases from leading humanoid manufacturers, we estimate that around 15k humanoid units were installed globally in 2025, a sharp jump from just 2k in 2024, and pointing to as many as 60k deployments in 2026, according to current company guidance (Figure 3).

² See International Federation of Robotics' [press release](#) (2023 data) and [press release](#) (2016 data).

FIGURE 3. The humanoid deployment era started in 2025

Note: 2026 data shows the consensus estimate.
Source: Company Reports, AlphaSense, Barclays Research

FIGURE 4. Chinese companies delivered 85% of global humanoid installations

Note: Based on 2025 data.
Source: Company Reports, Barclays Research

To date, deployment has been dominated by China and the US – with China clearly in the lead, accounting for nearly 85% of global installations in 2025. While installations in China have so far been concentrated among a small number of manufacturers, notably Unitree and AgiBot (Figure 4), this reflects the early scale-up phase rather than a narrow ecosystem.

In fact, China now hosts a rapidly expanding humanoid ecosystem, with roughly 140 active domestic manufacturers and around 330 humanoid models unveiled in 2025 alone.³ Crucially, this leadership did not emerge overnight. Our analysis in the next section shows that it is the result of more than a decade of sustained state-guided investment in technology, innovation, and talent, with a clear turning point around 2016.

China is the world's robotics powerhouse

Insights from 300k robotics patents

We use patent data to compare innovation trends in robotics across countries. Patents are inherently forward-looking signals of technological intent, revealing where firms expect future value to emerge, even if not all inventions ultimately reach commercial scale. Firms typically patent ideas they believe are both novel and economically viable, which means patenting activity tends to track the underlying R&D effort closely. This link is particularly strong in robotics, where innovation is incremental, highly technical and capital-intensive, and where patents often appear years before large-scale deployment.

Our analysis draws on [PATSTAT](#), the global patents database compiled by the European Patent Office (EPO). PATSTAT is one of the most comprehensive patent sources worldwide, covering close to 140mn patent filings. It provides detailed information on filing dates, technology classifications, applicants and geographic attribution, and is widely used in academic and policy research.

The key advantage of PATSTAT is that it harmonises patent information across jurisdictions and links related filings to a single underlying invention, avoiding double counting across countries. For example, a robotics invention may be filed first in the US and later extended to Europe and China as part of an international protection strategy. Treating these as separate filings would

³ See [Jamestown](#), and [The Guardian](#).

count the same invention multiple times, whereas an invention-based approach records it only once.

In our analysis, we date each invention using the earliest filing year, which captures when it first entered the patent system, and assign geographic origin based on the authority of the initial application.⁴ We focus on filings rather than grants, as grant timing varies widely across patent offices and often reflects procedural delays rather than the timing of the inventive activity.

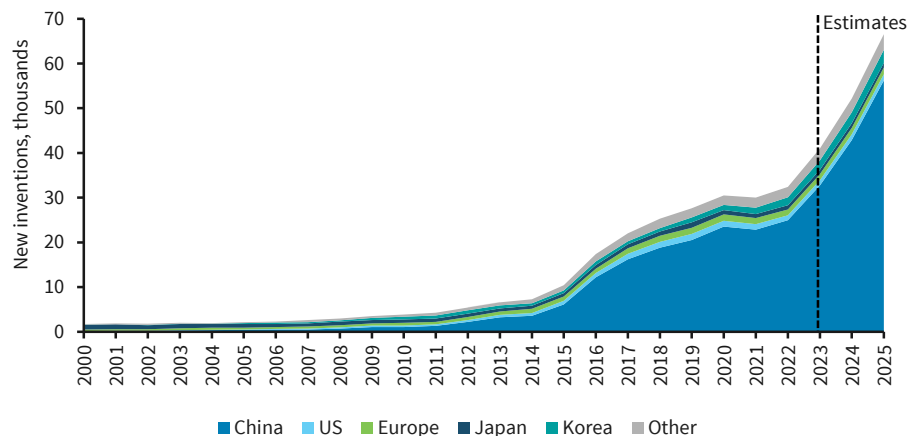
Our patents sample covers the broader robotics ecosystem, not just humanoids. Therefore, it captures both full-robot inventions and core component technologies – such as actuators, batteries, and motion systems – that are used across multiple physical AI robot types and can be integrated into humanoids. This broader framing helps avoid missing important sources of innovation that would be overlooked by focusing only on full humanoid patents. In PATSTAT, we identify robotics patents by selecting categories that include industrial robots, humanoids and drones. This yields a sample of over **300k robotics patents** since 2000.⁵

Invention counts after 2023 are affected by reporting and classification lags that tend to understate recent activity. This is because patents generally appear in PATSTAT 1–3 years after filing, once patent applications are published, leaving the most recent years structurally incomplete – a limitation that is well recognised in the literature.⁶ To address this effect, we combine available post-2023 data with the average growth rate observed over 2015–2023 to estimate invention activity in the most recent years.

Robotics innovation has shifted to China

Our analysis reveals that China accounts for 70% of global robotics inventions since 2000 (Figure 5). For example, in 2023 – the latest year with largely complete data – China recorded over 30k robotics patent inventions, almost thirty times higher than the number recorded in the US, which registered just over 1k inventions.

FIGURE 5. China accounts for nearly 70% of robotics inventions



Source: PATSTAT Autumn 2025, Barclays Research

This accelerating pace of innovation is also reflected in China's expanded domestic production of both components and finished robots. China now produces around 57% of the industrial

⁴ This reflects common practice in the literature, where the location of the first filing is used as a practical and consistent proxy for the origin of inventive activity.

⁵ Specifically, we use IPC subclasses B25J (industrial robotics and humanoids), and B64U (drones).

⁶ The [European Commission](#) and the [OECD](#) both highlight the impact of reporting delays on recent patent data.

robots installed domestically ([Figure 6](#)), marking a clear shift from the past, when installations relied far more heavily on imports, particularly from Japan.⁷

FIGURE 6. Today, most industrial robots in China are locally produced



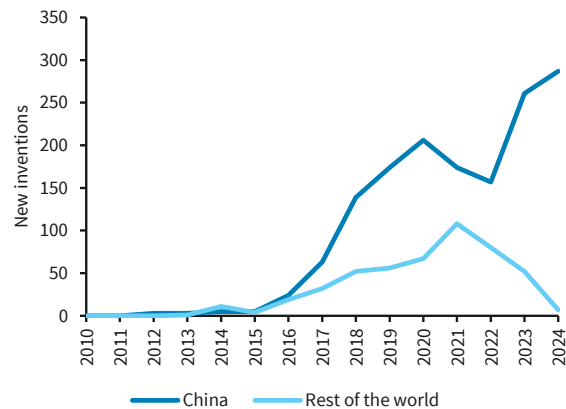
Source: International Federation of Robotics

While we view the broader robotics ecosystem as more informative, the results are very similar when the analysis is narrowed to humanoid-related patents – **with China accounting for about 75% of humanoid patents since 2010**. These results are based on patents filed by leading humanoid-robot manufacturers identified in our previous report ([The future of work: AI gets physical](#), 14 January 2026), which form the basis of our humanoid invention sub-sample. The sub-sample covers more than 20 firms behind the most advanced humanoid models as of late 2025 and, while excluding newer entrants yet to deploy, captures the companies responsible for the vast majority of humanoid units deployed globally to date.

Our analysis of humanoid patents shows that innovation in this space began to accelerate around 2015, with China already pulling ahead by 2018 ([Figure 7](#)). Much of this innovation momentum has been driven by a small group of firms, with UBTECH leading in China in terms of patenting activity and Boston Dynamics occupying a similar position in the US ([Figure 8](#)).

⁷ See International Federation of Robotics' [press release](#) (page 16).

FIGURE 7. China accounts for 75% of humanoid inventions



Source: PATSTAT Autumn 2025, Barclays Research

FIGURE 8. Leading innovators in humanoid robotics are based in China

Company	Country	Inventions
UBTECH	China	812
Xiaomi	China	239
Naver Labs	South Korea	134
Efort	China	155
Fourier Intelligence	China	132
Boston Dynamics	USA	119
Agibot	China	101
Shanghai Electric	China	63
Sanctuary AI	Canada	56
Leju Robotics	China	55

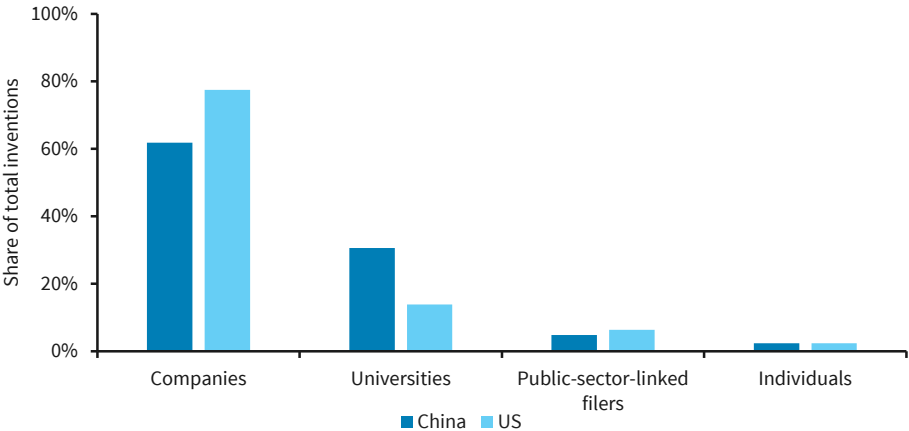
Note: Inventions column shows cumulative invention filings since 2010. We show data for the top 10 innovators.

Source: PATSTAT, Barclays Research

China leads robotics innovation via a university-government model

China's robotics machine is anchored in public research. Our analysis shows that universities and state research institutes account for over 30% of robotics patent inventions, versus around 10% in the US – reflecting an industrial policy strategy, formalised in initiatives such as *Made in China 2025*, that places academia at the core of technology development (Figure 9).

FIGURE 9. One-third of China's robotic inventions come from universities and public entities



Note: Based on 2023 data.
Source: PATSTAT Autumn 2025, Barclays Research

Software vs hardware: the US-China divide

Using PATSTAT data, we find a clear divide in robotics patenting – **China's patents focus on robot kinematics**, or the core physical structure, while **US patents are more concentrated on control and autonomy**, which govern how robots sense, decide and act.

We derive these results from detailed PATSTAT IPC classifications within our robotics patent sample. Because a single patent can span several technological applications, we allow each patent to contribute to multiple categories, reflecting the fact that many inventions are inherently multi-use. We then identify the top 10 application areas for China and the US, group them into four broad use cases, and assess their relative importance within each country's

patent portfolio. Based on this breakdown, we construct a patent use-case index, where a value of one corresponds to the dominant use case and other use cases are scaled proportionally. This approach shows a clear divergence: robot kinematics is the dominant use case of patents in China, with roughly twice the weight of control and autonomy applications, whereas the US profile is more tilted towards software-driven capabilities (Figure 10).

FIGURE 10. Robotics patent use-case index

Use case	China	US
Robot kinematic components	1.00	0.66
Safety & operational assurance	0.54	0.17
Control & autonomy	0.51	1.00
Manipulation & grasping	0.64	0.27

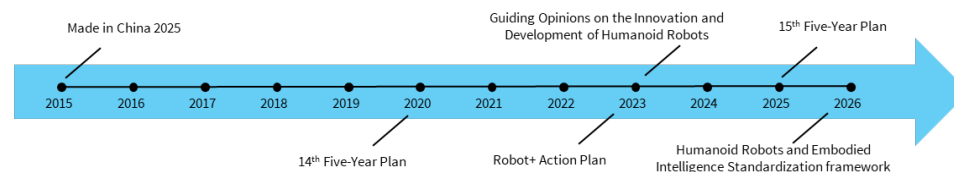
Source: PATSTAT Autumn 2025, Barclays Research

In China, robotics is industrial policy

State policy has underpinned China's robotics lead

The expansion of robotics deployment and innovation in China is not surprising. In our view, **China's robotics expansion stems from a long-running, government-coordinated industrial policy framework**, which began to accelerate in the mid-2010s. These policies are embedded in a series of national planning documents, beginning with *Made in China 2025*, issued in 2015, and followed by industrial strategies under the 14th and 15th Five-Year Plans (Figure 11). Together, they designated robotics and embodied intelligence as strategic-infrastructure industries and shaped their development by enabling the coordinated use of policy tools, including subsidies, grants, tax incentives, and state-backed financing at both central and local levels.

FIGURE 11. China's core robotics policy framework



Source: Barclays Research

More recently, the 2023 Robot+ Action Plan set out guidelines for the adoption and integration of robotics across the broader economy, identifying 10 priority industries for deployment (Figure 12),⁸ signaling a second phase of policy focused on extending robotics into under-penetrated areas.

FIGURE 12. China's 10 key robotics application industries

Deployment industries	
Manufacturing	Healthcare
Agriculture	Elderly services
Architecture	Education
Energy	Business community services
Commercial logistics	Emergency applications

Source: Barclays Research

⁸ See CSIS China Power Project, *The Robot Report* and *bodk*.

Ahead of schedule

These policies are already paying off, with **several targets reached ahead of schedule**. According to the Center for Strategic and International Studies (CSIS), the *Made in China 2025* strategy set a target of reaching 1.8mn industrial robots in operational stock by 2025. This level was reached in 2023, with total stock exceeding 2mn in 2024. Likewise, key objectives under the 14th Five-Year Plan, including average annual growth of 20% in robotics revenues and a doubling of robot density per 10k workers by 2025, were also achieved earlier than initially envisaged.⁹

China is building the entire humanoid robotics stack

Meanwhile, Chinese state policy on humanoid robots treats them as a system, rather than a standalone project, with a focus on building the entire ecosystem. This **approach addresses both supply and demand**: from raw materials and components to humanoid producers, while simultaneously encouraging deployment to create domestic demand.

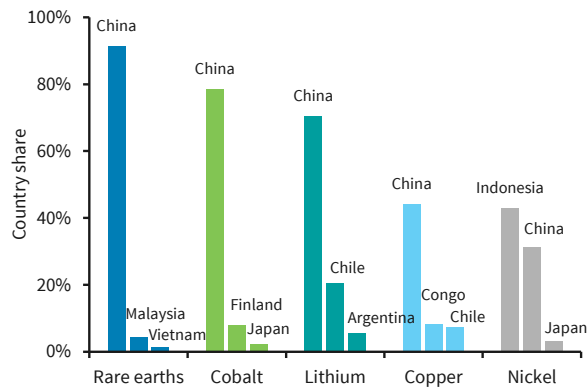
Vertically integrated supply chain

China's key advantage in deploying humanoid robots is its control over the full supply chain. From upstream raw inputs such as critical minerals to intermediate goods and final components, China has secured a tightly integrated and reliable production base. Crucially, the West remains highly dependent on this supply chain to meet their own robotics demand.

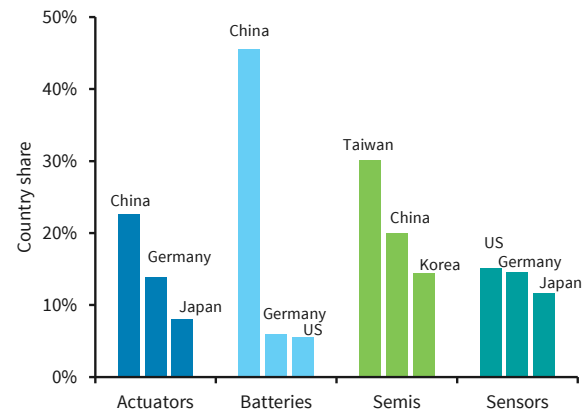
Our previous research shows that **more than 90% of global refined magnetic rare earths come from China** – materials that are essential for electric-based locomotion in robots ([Figure 13](#)). China is also the world's leading processor of refined cobalt, lithium and copper, and ranks second in nickel, all of which are critical inputs into robotic electronic systems, including batteries and wiring.

Beyond raw materials, China also **dominates the midstream and downstream manufacturing of humanoid components**. Our analysis of the UN Comtrade database – the most comprehensive source of detailed international trade flows – shows that China is the world's largest exporter of key intermediate inputs used in robotics and humanoid robots. Specifically, China accounts for more than 22% of global actuator exports (rare-earth-powered devices that convert electrical energy into controlled mechanical motion), 45% of battery exports, and is the second-largest exporter of semiconductors ([Figure 14](#)).

⁹ See [CSIS China Power Project](#).

FIGURE 13. China is the top supplier of refined critical minerals

Note: 2024 data. Shares reflect each country's share of global supply.
Source: International Energy Agency Licence: CC BY 4.0, Barclays Research

FIGURE 14. China also leads global exports of critical components

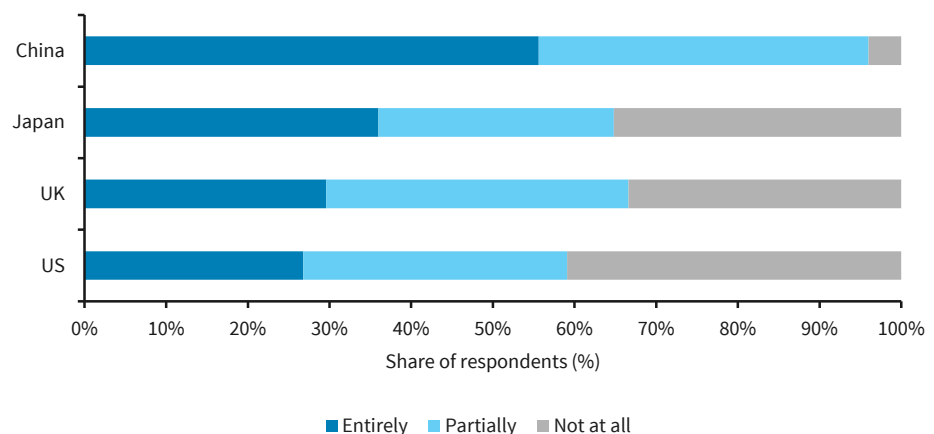
Note: 2024 data. Shares reflect each country's share of global exports.
Source: UN Comtrade, Barclays Research

More robots → more data → better robots → more robots

With nearly 13k humanoid robots deployed in China by end-2025, early scaling is already generating powerful second-order effects. Larger deployments are producing growing volumes of real-world operating data, which directly improve humanoid robot models – helping close the training gap left by simulations and data drawn from images or video alone. Data sharing and standardisation amplify these gains.

Social acceptance provides an additional tailwind: surveys by the Mitsubishi Research Institute indicate that close to 60% of Chinese respondents view robots favourably in consumer services, a much higher acceptance rate than in the US and the UK (Figure 15).¹⁰

While deployment is likely to remain predominantly industrial over the next 5-10 years, this openness lowers adoption frictions and creates a self-reinforcing loop: deployment generates data, data improve performance, and improved performance accelerates further deployment.

FIGURE 15. China shows the highest public acceptance of robots

Note: Data shows desired degree of robot involvement in consumer services.
Source: Mitsubishi Research Institute

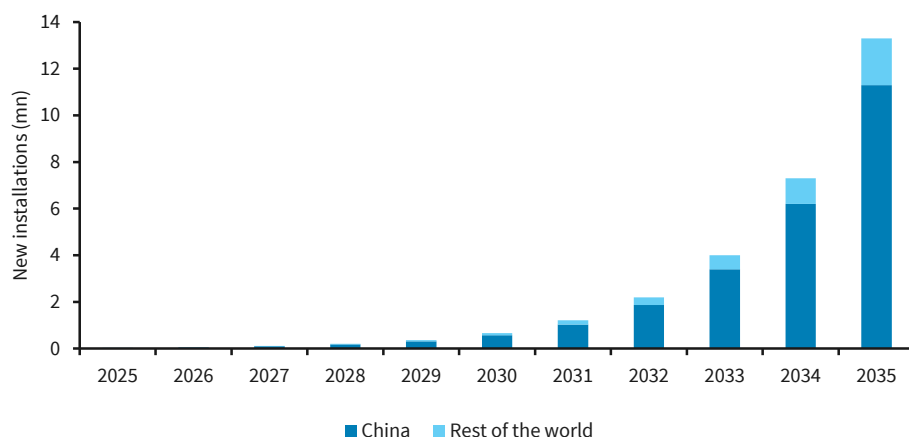
¹⁰ See Mitsubishi Research Institute's [press release](#).

Against this backdrop, the policy objective of cultivating 2-3 globally influential enterprises by 2025 appears aligned with industry progress, with AgiBot and Unitree already emerging as leading global producers – pointing to meaningful advances toward a competitive industrial ecosystem by 2027.¹¹

Robots, made in China

Under an upper-bound scenario, global humanoid robot production could reach up to 13mn units a year by 2035, with around 85% – concentrated in China (Figure 16). Early deployment is expected to be largely shaped by supply-side constraints, as manufacturing capacity ramps up, which is why we first focus on production before examining demand considerations in the following chapter.

FIGURE 16. China is poised to lead global humanoid robot deployment through 2035



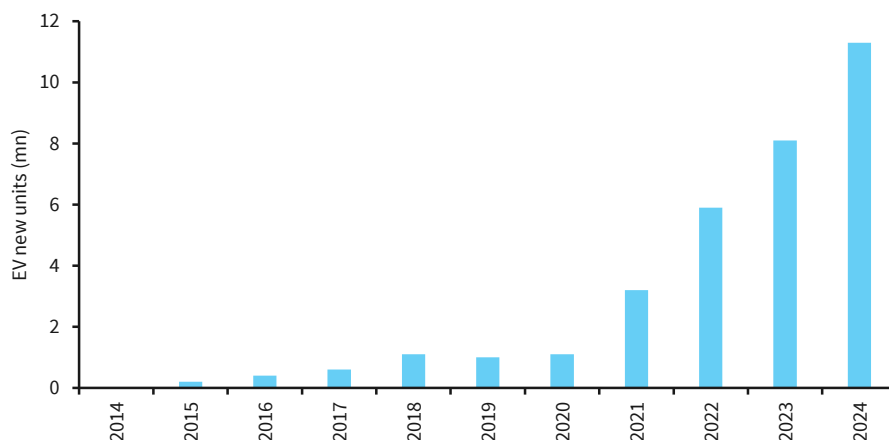
Source: Barclays Research

Humanoids are (electric) cars in miniature

At this early stage, most bottom-up forecasts for humanoid robots rest on highly uncertain assumptions around unit costs, utilisation, returns on investment and regulatory acceptance.

We assess humanoid manufacturing potential in China by benchmarking it against the roll-out of China's electric vehicles (EV) expansion in the 2010s, which we view as the most comparable technology (Figure 17). The EV experience offers a relevant, real-world example of how quickly a complex, capital-intensive technology can scale once adoption inflects. While humanoids and EVs clearly serve different end-markets, EVs provide a relevant analogue for manufacturing scale, as both face very similar industrialisation challenges. In particular, both are built on the same core components – semiconductors, actuators, gears and bearings, and batteries (Figure 13 and Figure 14) – and similarly reliant on large-scale systems integration and learning-by-doing. As with EVs in their early years, the humanoid industry is currently heavily concentrated in China and supported by active government backing, making the EV adoption curve a particularly relevant and realistic reference point. We stress that this analysis should be interpreted as an **upper bound scenario** rather than a precise forecast.

¹¹ See [MIT China](#), [Global Times](#) and [Robot Today](#).

FIGURE 17. China's EV sales have seen exponential growth

Source: International Energy Agency Licence: CC BY 4.0

Based on these similarities, we project China's humanoid deployments by mapping adoption onto China's historical EV roll-out following its inflection point. We treat 2025, with roughly 15k humanoid deployments, as year one. Then, we anchor 2035 annual humanoid deployments in China at around 11mn units – the level of EV new sales reached roughly 10 years after EV adoption began to accelerate (Figure 17). Finally, we align 2026 humanoid deployments with the industry consensus estimate of 60k units, and interpolate deployment volumes over 2027–35 using the implied compound annual growth rate consistent with this endpoint for tractability, acknowledging that realised adoption is likely to be uneven rather than smooth.

We then translate this China-level projection into a global estimate by assuming China retains an 85% share of worldwide humanoid deployments, reflecting its current leadership in manufacturing and concentration of early adoption in industrial use cases, while recognising that this share could decline over time as production and deployment diversify geographically. Under this assumption, total global deployments would reach up to **13mn units by 2035 – roughly up to 11mn in China and up to 2mn outside China.**

If current trends continue to hold, we expect the majority of these non-China deployments to be produced by **Tesla, Figure AI, Boston Dynamics** and **Agility** (in the US) and by Sweden's **Hexagon AB**. Of course, at face value, producing up to 2mn units outside of China appears very ambitious. That said, this scale is broadly consistent with the production targets articulated by several of the most prominent industry leaders in the space.

In its fourth-quarter earnings call in late January, Tesla announced plans to end production of its flagship Model S and Model X vehicles as it shifts towards lower-cost models, while converting the Fremont plant – previously used to manufacture these vehicles – into a humanoid robot production site. Notably, under the terms of Elon Musk's recently approved compensation package, which is tied to ambitious valuation and product roll-out milestones, Tesla's CEO would need to deliver roughly 1mn humanoid robots by 2035.

Boston Dynamics targets manufacturing 30k Atlas humanoid units annually by 2028¹². Figure AI has publicly targeted mass production of 100k humanoid robots within a similar time frame¹³ and Agility's existing RoboFab facility could scale up to 10k robots annually at peak production¹⁴. Hexagon AB, albeit taking a more cautious approach to explicit targets, has

¹² See Boston Dynamics' [press release](#).

¹³ See Figure AI's [press release](#).

¹⁴ See Agility's [press release](#).

indicated plans to produce at least a few thousand units by early 2030. Including smaller volumes from other producers, this points to aggregate production of roughly 200k units by 2028-29. If production inflects exponentially from around 2030, as many industry participants expect, output could then ramp up materially.

Why smartphones and industrial robots are less appropriate analogues

Smartphones are a mass-market consumer good, whereas humanoid robots are likely to remain predominantly industrial assets over at least the next decade, with adoption driven by enterprise deployment rather than personal ownership.

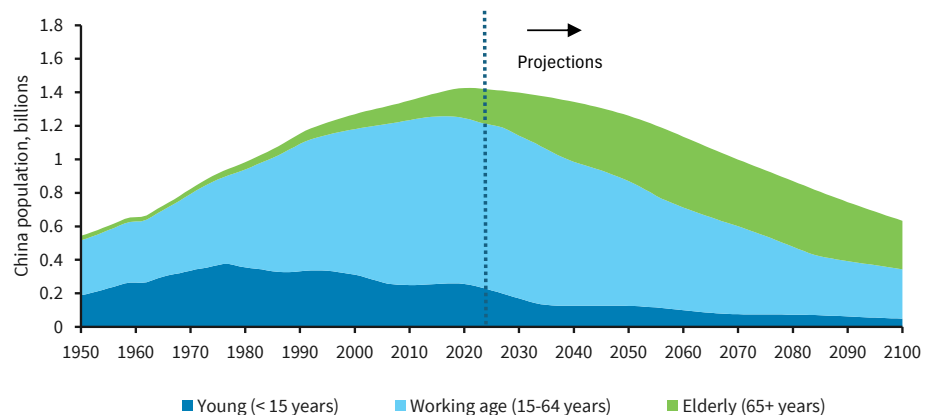
Industrial robots would also be a poor analogue because they automate processes whereas humanoids automate labour. Put differently, one scales with tasks, the other scales with workers. Industrial robot demand is ultimately bounded by the number of discrete, automatable processes within a factory – once each task is automated, incremental demand slows sharply. Humanoids, by contrast, are designed to substitute for or augment human workers across a wide range of tasks, environments and sectors. Their deployment therefore scales with the size of the labour force, not with the number of production processes. This distinction materially expands the addressable market and implies much higher potential unit volumes than those observed for traditional industrial robotics.

Robots against demographics

China is aging, fast

China's demand for robots is rising as demographic pressures build from a shrinking and aging population. The demographic shift is already underway, China's population peaked at just over 1.42bn in 2021 and is now declining, with projections pointing to around 1.2bn by 2050 (Figure 18). These pressures will be felt acutely in the labour market, with **working-age population in China projected to fall by 6% between 2026 and 2035, and by 25% by 2050.** Our analysis of demographics data suggest that this would equate to roughly 57mn fewer working-age people, which, if the participation rate remains at 65%, would translate into a 37mn decline in the labour force within the next decade alone. At the same time, the share of elderly in the population is projected to rise from about 15% to over 30%, intensifying demand for healthcare and elderly care workers.

FIGURE 18. China's population has already peaked



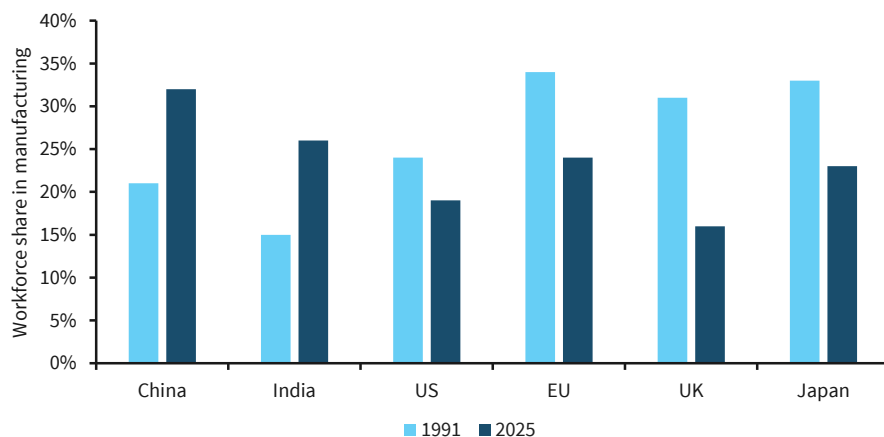
Source: United Nations' World Population Prospects 2024, Barclays Research

Against this backdrop, **shifts in population dynamics pose growing challenges for China's labour-intensive manufacturing model.** As work preferences continue to tilt from industry toward services, labour constraints are likely to become more binding for sustaining the world's

largest industrial base – valued at around \$4.7trn as of 2024¹⁵ and accounting for roughly 25% of China's GDP. Manufacturing still employs about 245mn workers, or roughly 32% of the workforce (Figure 19), far higher than in most large economies that have already transitioned further toward services.

As China's population shrinks, part of the potential output loss could, in principle, be offset by gains in total factor productivity (TFP). However, despite strong historical performance, projections from the UN suggest that productivity growth is likely slow to around 1% a year, implying only around 11% cumulative TFP growth between 2026 and 2035.¹⁶ At this pace, efficiency gains would offset only a small share of the projected labour force decline, and are unlikely to fully counter China's demographic headwinds – thereby strengthening the **economic case for automation and robotics as a partial offset to tightening labour supply**.

FIGURE 19. Workforce share in manufacturing



Source: World Bank, Barclays Research

Can robots offset demographic tailwinds in China?

Humanoid deployment in China could plausibly complement productivity gains in mitigating labour force losses over the coming decade. Under optimistic scenarios, and if current exponential scale of innovation and deployment continue to hold, the cumulative installed stock of humanoid robots in China could approach around 24mn units by 2035. This estimate reflects the cumulative sum of projected annual deployments between 2031 and 2035 (Figure 16), assuming that robots installed prior to 2031 are fully depreciated by then, and that **domestic production is fully deployed within China** (see next section for a more robust discussion). This appears plausible, as workers retiring in China over the 2030s are likely, on average, to have lower educational attainment and greater exposure to manual tasks – categories in which humanoid robots may offer the greatest scope for substitution.

At this scale, humanoid robots could be equivalent to up to **3.8% of China's labour force by 2035**, corresponding to an additional labour input on the order of up to 24mn workers, or roughly 60% of the projected demographic shortfall. We stress that this estimate should be viewed as an upper bound, reflecting relatively optimistic assumptions around utilisation, depreciation and domestic absorption.

¹⁵ See [World Bank](#).

¹⁶ Penn World Table, IMF, Barclays Research

The US and Europe are still in the race

Applying the same upper-bound scenario analysis as for China suggests the **US could have a stock of up to 3.7mn humanoid robots by 2035, equivalent to up to 2.5% of the labour force**. At face value, this points to weaker adoption in the US, but we do not view this as an apples-to-apples comparison. Rather, it reflects differences in deployment profile, timing and use-case mix.

In the US, humanoids go consumer – but much later

Our analysis suggests that humanoid deployment over the next five to seven years is likely to be concentrated predominantly in industrial applications, particularly manufacturing. China's manufacturing sector accounts for around 25% of GDP, underscoring the scale of its industrial base. By contrast, the US is primarily a services-oriented economy, with manufacturing accounting for roughly 9% of GDP. As a result, near-term demand for humanoid robots in China is likely to be materially larger in absolute terms than in the US, reflecting the much greater scale of its manufacturing sector. We expect deployment in the US to be driven more by service-oriented applications, with adoption accelerating in the 2030s as these use cases mature. Moreover, the US faces no immediate pressure to offset labour-force decline, as UN projections indicate continued labour-force growth, reaching around 141mn by 2035, up from 139mn in 2024.

Safety and security create a window for Western producers

To be clear, we are not arguing that the US or Europe are currently ahead – China's advantage at this stage is sizable. However, we do see a meaningful opportunity for the West to narrow the gap over time. While Chinese manufacturers are likely to retain a cost and pricing advantage, we expect safety, security and cybersecurity considerations to remain central for companies looking to deploy humanoid robots. These systems are highly sensor-rich, and the risks around data leakage, intellectual property loss, or **embedded spyware** are nontrivial.

As a result, we expect regulators in both the US and Europe to introduce robust guardrails governing which Chinese technologies can be imported and deployed, particularly from a national security perspective. **This informs our baseline assumption that a significant share of China's humanoid robot production will ultimately be deployed domestically.** Against this backdrop, with renewed emphasis on industrial policy in the West, we see an opening for humanoid producers outside China to step in and scale.

A wave of M&A activity

In the West, our analysis suggests that Tesla, Figure AI and Agility in the US, alongside Sweden's Hexagon AB and South Korea's Boston Dynamics, are currently among the leaders. At the same time, the Western humanoid robotics ecosystem is still unfolding, with a growing number of companies entering the space and experimenting with different technologies, form factors and use cases. We argue, however, that scale will be critical. Many of these firms are small, software-first, and constrained by limited manufacturing capacity. As a result, we expect consolidation to play a key role, with **increased M&A activity in the next three to five years likely across Europe and the wider West** as the ecosystem matures.

Autonomy meets geopolitics and industrial policy

One stack, many frontiers

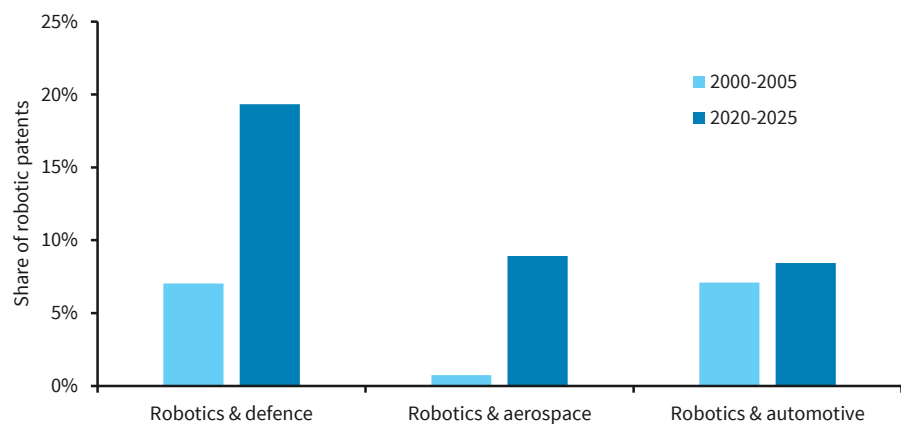
Current deployment and invention activity in robotics is likely to generate important second-order effects beyond its initial use cases, most notably in defence, energy and other frontier systems such as space technology. These spillovers are driven less by the end products

themselves than by **shared foundational technologies** – including autonomy and control, navigation, sensing and perception, actuation and mobility, and power systems.

Our patent analysis identifies **three industries where robotics spillovers are becoming increasingly significant: defence, aerospace, and automotive**. We identify these spillovers by tracking patents that sit at the intersection of robotics and industry-specific technological classifications. Because patents can be assigned to multiple classifications rather than a single end use, this approach allows us to observe growing application overlap across sectors.

The strongest convergence is between robotics and defence. Nearly 20% of robotics patents filed between 2020 and 2025 are also classified under defence-related technologies, up from around 7% in the early 2000s (Figure 20). Spillovers into aerospace are smaller but rising, with roughly 9% of robotics patents also linked to aerospace applications. Overlap with the automotive industry has historically been high, but we observe a further increase even there.

FIGURE 20. Robotic patents overlap with defence, aerospace and automotive has grown in recent years



Source: PATSTAT Autumn 2025, Barclays Research

Implications for investors and lessons from China

1. **Opportunities among the technology stack, not just the end product.** Our analysis suggests that returns will likely concentrate in five foundational technologies: autonomy software, sensing, actuation, batteries and edge controls. These capabilities are reusable across robotics, defence, energy and space, expanding the addressable market. For example, solid-state batteries extend humanoid run-time, underpin grid-scale power storage, enable longer-range drones and extend satellite life due to their lighter weight and radiation tolerance. Suppliers of components and enabling technologies therefore benefit from multiple, overlapping adoption waves, reducing reliance on any single end market or deployment timeline.
2. **Defence and frontier demand provide downside protection.** Spillovers into defence energy and even potentially space tech introduce non-cyclical demand sources. This can stabilise revenues and accelerate scale, especially where public procurement underwrites early adoption.
3. **Many-use, not dual-use.** As AI becomes more deeply embedded in modern economies, we are likely to see a shift in how investment themes are framed – away from allocating capital purely by industry (such as technology, healthcare, or defence) and towards investing across the technology stack. This cuts across how markets and governments are organised today – investors allocate capital by sector, and public spending is likewise structured along sectoral lines. Physical AI does not sit neatly within this framework. We argue that scale is

unlikely to come from a single, viral application, but rather from the aggregation of multiple use cases deployed across a wide range of industries.

Implications for policy

1. **Industrial policy shifts from backing strategic industries to securing control over strategic technologies.** Rather than treating robotics as a standalone sector, policy focus is increasingly turning to control over the underlying technology stack. Looking ahead, strategic autonomy is likely to mean treating the robotics supply chain as critical infrastructure.
2. **Supply-chain security now matters more than final assembly.** Strategic vulnerability increasingly sits in components, inputs and software, not finished products. From a resilience perspective, our analysis identifies key choke points – most notably critical minerals, batteries and actuators (Figure 13 and Figure 14) – which are emerging as a new source of geopolitical leverage.
3. **Demand resilience matters as much as supply resilience.** China already deploys a large number of robots domestically and, in our view, is likely to continue doing so. Deployment creates domestic demand not only for the end product, but for the entire upstream robotics ecosystem. This supports the development of end-to-end solutions that span every stage of the supply chain – a dynamic that the US and other Western economies will need to factor into their own industrial and technology strategies.

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